GRAPHICS CARDS

2017 should be a momentous year

This was a year of much joy for the PC Master Race. Both companies, NVIDIA and AMD made huge improvements to their GPU architecture. The move to 16 nm and 14 nm FinFET design has obviously worked wonders for NVIDIA's Pascal and AMD's Polaris architectures. NVIDIA went about the usual task of releasing new graphics cards based on Pascal across the entire price segment with the GP107 based 1050 at one end targeted at low to mid-range gaming and the GP102 based TITAN X at the other end for GPGPU applications. On the other hand, AMD's Polaris was only put into production in the low to mid-range segment with the Polaris 10 and 11 GPUs in the AMD RX 460, RX 470 and RX 480. While questions were raised about this approach, it did work wonders for AMD whose shares rose from $1.67 (Jul '15) to 8.7 (November '16). The entire GPU market sold a lot more units compared to the previous year with NVIDIA taking the lead followed by AMD and Intel.

And when AMD's VEGA GPU does launch for the high-end segment we'll get to see some real action.

ZERO1 WINNER

NVIDIA GeForce GTX 1080 - GIGABYTE

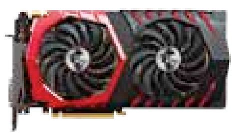
The new TITAN X is not available in India. Which leaves the GTX 1080 as the victor. AMD has no GPU to compete against the likes of the GTX 1080 at the moment (though we hope VEGA comes out at CES if not around the rumoured mid-December launch date). The ASUS ROG STRIX GTX1080 O8G GAMING and the GIGABYTE GTX 1080 XTREME GAMING PACK were neck and neck given they both had the exact same clock speeds and all our benchmark scores were within 1% or each other so we couldn't decide between them easily.

The ZOTAC and the MSI cards are only ever so slightly clocked lower than the other two and that brings them outside the 1% error margin that we follow. So that brings us back to the GIGABYTE and the ASUS cards. Looking at the cooling performance, we did notice that the ASUS card had slightly higher temperatures on full load. We suspect this could have been due to the fact that the ASUS card came to us much later so every other reviewer who had it before us could have possibly disassembled the card.

NVIDIA GeForce GTX 1080 - GIGABYTE
Price: ₹66,250

ASUS GeForce GTX 1080
Price: ₹64,000

ZOTAC GeForce GTX 1080
Price: ₹58,873



MSI GeForce GTX 1080
Price: ₹63,000

BEST BUY digit DECEMBER 2016

AMD RX 480 - Sapphire
Price: ₹22,999

We don't overclock the cards for Zero1 but if we are to consider the fact that the GIGABYTE card has a 12+2 VRM design as compared to the ASUS' 8+2 design, then theoretically, the GIGABYTE will have a more stable current allowing for stable overclocks. But more isn't always better. However, considering the fact that gaming performance is the same, we resort to temperature performance being the deciding factor to tip the scale and that makes the GIGABYTE GTX 1080 EXTREME GAMING PACK the Zero1 Winner.

RUNNER-UPS

ASUS, ZOTAC and MSI GeForce GTX 1080

ASUS misses out by a sliver and the ZOTAC and MSI GeForce GTX 1080 cards end up as the runner-ups in the graphics segment. All three cards have high over-clocks, good accompanying software, and beefy cooling solutions. With each card being so close to each other in performance, it becomes really difficult to pick between the three. So when you bring cost into the picture and their respective warranty periods then the ZOTAC card shines. The ASUS card just narrowly lost out winning the Zero1 Award and the MSI card has one of the best build quality in the premium segment. So make your choice!

BEST BUY

AMD RX 480 - Sapphire

The AMD RX 480 gets the Best Buy because of a multitude of factors. While NVIDIA's GTX 1060 and RX 480 are neck and neck in most games with either card taking the lead in games that use proprietary APIs. However, given that the AMD card performs better in games using Directx 12 and Vulkan APIs, we have to give it to AMD. Barring the initial pricing goof-up, the RX 480 is now at a much better price point.